The map sending a matching to its core and its standard Young tableau is well defined

Avichai Marmor

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Editorial note. The counted statement is the author's unlabelled corollary environment at publisher-source.tex lines 854-860; its proof environment opens at line 861, immediately after \end{corollary}, so the pairing is by adjacency and the retained proof range 861-869 is the whole environment.

Editorial note. The proposition prop:core well defined, statement and proof (source lines 846-853), is retained as context because the counted proof cites it at source line 862; it is the only outside label of the retained proof.

Editorial note. Because the corollary carries no author label, its identity is recorded by the pinned source position of the two environments (statement lines 854-860, proof lines 861-869) and by the claim key, which encodes that position.

Editorial note. This article carries no label-number dump and the retained counter values are left empty for the whole source, so the printed numbering of the card is the wrapper's default; the original printed number of this unlabelled corollary is not recoverable from the pinned material and is therefore not asserted.

Editorial scope. One original result of the pinned publisher article, counted once: well-definedness of the map from a matching to its core and its two-row tableau (unlabelled corollary at publisher-source.tex lines 854-860).

Allowed standard prerequisites (editorial).

- The union of cores and two-row tableaux. The target is the disjoint union over k of the matchings with short-chord-free core and the standard Young tableaux of shape (N-k,k).. It is the codomain in which the map F is asserted to be well defined.
- Core and two-row tableau of a matching. The core is the matching on the stable vertices and T(m) is the two-row tableau whose second row records the unstable chords.. They are the two components of the map.
- Standard Young tableau. A two-row tableau of shape (N-k,k) is standard when both rows and both columns are increasing and each letter occurs once.. It is the property the proof verifies for T(m).

Proposition 0.1. Given a matching, its stable vertices and its core are well-defined.

Proof. Let $m \in \mathcal{M}_N$ be a matching. Denote the chords that are removed during a given reduction process of m by $e_1, \ldots, e_k$, and denote $e_\ell = (i_\ell, j_\ell)$. Moreover, denote the chords that are removed during another reduction process by $e'_1, \ldots, e'_{k'}$, and denote $e'_\ell = (i'_\ell, j'_\ell)$. It suffices to prove that $e_\ell \in \{e'_1, \ldots, e'_{k'}\}$ for all $1 \leq \ell \leq k$ (i.e. every chord that is removed during the first reduction process is removed during the second process as well). Assume by contradiction that $e_\ell \notin \{e'_1, \ldots, e'_{k'}\}$ for some ℓ, and denote by ℓ_0

the minimal such ℓ. That is, $e_1, \ldots, e_{\ell_0-1} \in \{e'_1, \ldots, e'_{k'}\}$. Removing the chords $e_1, \ldots, e_k$ from m constitutes a valid reduction process, so the chord e_{ℓ_0} becomes short before it is removed. That is, $i \in \{i_1, j_1, \ldots, i_{\ell_0-1}, j_{\ell_0-1}\}$ for all $i_{\ell_0} < i < j_{\ell_0}$. Therefore, the chord e_{ℓ_0} is a short chord of the core obtained by removing $e'_1, \ldots, e'_{k'}$ of m, contradicting the requirement that the reduction process continues until there are no short chords remaining. Therefore, we may conclude that if a chord is removed during a reduction process then it is removed during any reduction process, and $\mathrm{core}(m)$ and $\mathrm{Stable}(m)$ are well-defined. □

Corollary 0.2. Let N and f be nonnegative integers. Then the function

$$F : \mathcal{M}_{N,f} \to \bigcup_{k=0}^{n} \mathcal{M}_{N-2k,f}(\mathrm{Short} = \emptyset) \times \mathrm{SYT}(N-k, k)$$

defined by $F(m) = (\mathrm{core}(m), T(m))$ is well-defined.

Proof. Let $m \in \mathcal{M}_{N,f}$ be a matching. By Proposition 0.1, we obtain that $\mathrm{core}(m)$ and $\mathrm{Stable}(m)$ are well-defined. Therefore, the two-row tableau $T := T(m)$ consisting of N cells that is defined by

$$\mathrm{row}_2(T) = \{j \mid \text{the chord } (i, j) \in m \text{ is unstable}\}$$

is also well-defined. In order to prove that F is well-defined, it remains to show that $T(m) \in \mathrm{SYT}(N-k, k)$, where k is the number of unstable chords of m.

Obviously, every letter $i \in [N]$ appears exactly once in $T(m)$, and the rows are increasing. Notice that for every entry $j \in \mathrm{row}_2(T(m))$ there exists $i < j$ such that $(i, j) \in m$ is an unstable chord. Therefore, every entry $j \in \mathrm{row}_2(T(m))$ is associated to an entry $i \in \mathrm{row}_1(T(m))$ such that $i < j$. Moreover, if $j \neq j' \in \mathrm{row}_2(T(m))$ and $(i, j),\ (i', j') \in m$ then $i \neq i'$, so the columns of $T(m)$ are increasing and $T(m)$ is a standard Young tableau. Finally, $|\mathrm{row}_2(T(m))| = k$, implying that $T(m) \in \mathrm{SYT}(N-k, k)$. □